ATOM COUNT CARD

Cobra Energy Drink: from a printed label to moles, molecules and atoms

Student: Maryan M. Morillo · Chemistry for Engineers · Asynchronous Activity 02 · Mole Concept & Mole Conversions

A. Product Identity



Product: 330 mL bottle. Label: 240 mL serving, 1.4 servings per bottle
Compound: sucrose, $C_{12}H_{22}O_{11}$ (from "Sugars 23 g" on the label; the ingredient list names "Sugar").
Why it matters: sugar is the main energy source in the drink.
Student: Maryan M. Morillo, Chemistry for Engineers.

B. Base Quantitative Data

Quantity	Value	Source	Precision
Sugars	23 g per serving	Product label	2 s.f. (limiting)
Servings per bottle	1.4	Product label	2 s.f.
Serving size	240 mL	Product label	Not limiting
Bottle volume	330 mL	Product label	Not limiting
Formula	C12H22O11	PubChem	Exact subscripts
Atomic masses C, H, O	12.011, 1.008, 15.999 g/mol	CIAAW	Not limiting

Assumption: "sugars" are treated as sucrose.

C. Molar-Mass Build-Up

Element	Atoms	g/mol each	g/mol	% by mass
C	12	12.011	144.132	42.107
H	22	1.008	22.176	6.479
O	11	15.999	175.989	51.414
C12H22O11	45		342.297	100.000

Addition rule: each contribution has 3 decimal places, so M keeps 3 decimal places. Percentages add up to 100.000 % (check).



Nutrition Facts

Serving Size: 240 mL
No. of Servings per bottle: 1.4

Amount Per Serving:		%RE/RNI*
Calories	90 kcal	4%
Calories from Fat	0 kcal	
Total Fat	0 g	
Saturated Fat	0 g	
Trans Fat	0 g	
Cholesterol	0 mg	
Sodium	50 mg	
Total Carbohydrates	23 g	
Dietary Fiber	0 g	
Sugars	23 g	
Total Protein	0 g	0%
Thiamine Hydrochloride	(Vit. B1) 1.4 mg	120%
Zinc	6.5 mg	100%
Cholecalciferol	(Vit. D3) 200 IU	100%
Niacinamide	(Vit. B3) 8.0mg	50%
Pyridoxine Hydrochloride	(Vit. B6) 0.8 mg	70%
Cyanocobalamin	(Vit. B12) 0.8 mcg	35%

*Based on Philippine Dietary Reference Intakes (PDRI) 2015 for adult male (19-29 years old)

D. Molar-Mass Build-Up

3.2×10^{1} g sucrose
23 g/serving × 1.4 servings

↓ ÷ 342.297 g/mol

9.4×10^{-2} mol sucrose

↓ × 6.022 140 76 × 1023 molecules/mol

5.7×10^{22} molecules sucrose

↓ × 12 C atoms / molecule

$.8 \times 10^{23}$ C atoms

E. Mole-Conversion Calculations

1. **Molar mass: 1**2(12.011) + 22(1.008) + 11(15.999) = 144.132 + 22.176 + 175.989 = 342.297 g/mol (3 decimals)
2. **Label bridge:** (23 g / 1 serving) × 1.4 servings = 32.2 g → 3.2×10^{1} g sucrose per bottle
3. **Mass → moles:** 32.2 g × (1 mol / 342.297 g) = 0.094070 → 9.4×10^{-2} mol sucrose
4. **Moles → molecules:** 0.094070 mol × ($6.022\,140\,76 \times 10^{23}$ molecules / 1 mol) = 5.6650×10^{22} → 5.7×10^{22} molecules sucrose
5. **Atoms of one element (C):** 5.6650×10^{22} molecules × (12 C atoms / 1 molecule) = 6.7981×10^{23} → 6.8×10^{23} C atoms
6. **Percent composition + two-route check: %C** = 144.132 / 342.297 × 100 = 42.107 %.
   Route A: 32.2 g × 0.42107 = 13.56 g C.
   Route B: 0.094070 mol × (12 mol C / 1 mol sucrose) × (12.011 g / 1 mol C) = 13.56 g C.
   Both routes → 14 g C
7. **From % composition back to the formula:** In 100 g: C 42.107 g ÷ 12.011 = 3.5057 mol; H 6.4786 g ÷ 1.008 = 6.4272 mol; O 51.414 g ÷ 15.999 = 3.2136 mol. ÷ 3.2136 → C 1.091 : H 2.000 : O 1.000 → ×11 → 12 : 22 : 11 → $C_{12}H_{22}O_{11}$ (empirical = molecular, n = 1)
8. **Scale-up and reverse:** 40 bottles × 0.094070 mol/bottle = 3.763 mol → 3.8 mol. Reverse: 1.00 mol × (342.297 g / 1 mol) × (1 bottle / 32.2 g) = 10.63 → 11 bottles per mole.

F. Scale-Up: from one bottle to a class

1 BOTTLE	1 CLASS: 40 BOTTLES	EXAM WEEK: 200 BOTTLES	REVERSE: 1 MOLE
9.4×10^{-2} mol	**3.8 mol**	**1.9×10^{1} mol**	**11 bottles**
5.7×10^{22} molecules	2.3×10^{24} molecules	1.1×10^{25} molecules	342.297 g ÷ 32.2 g per bottle = 10.63 → 11
3.2×10^{1} g sucrose	1.3×10^{3} g (≈ 1.3 kg)	6.4×10^{3} g (≈ 6.4 kg)	

Bottle counts (40 students; 5 exam days) are exact assumptions made for illustration. Results use the label's 2 s.f.

G. Sig-Fig Check + One-Number Takeaway

Calculator: 5.66505×10^{22} molecules → **Reported:** 5.7×10^{22} molecules
Limits: 23 g and 1.4 servings (2 s.f. each). **Exact:** N_A, subscripts, number of bottles.
Guard digits kept until the last line of every calculation; no rounding in between.

★ *One bottle of Cobra has about 1.2×10^{23} sucrose molecules, yet 5.0 bottles make only 1 mole (342 g).*

H. References (APA 7th) + AI Disclosure

Interbev Philippines, Inc. (n.d.). Cobra Energy Drink, 330 mL [Product label]. Manufactured at Km. 43 National Highway, Brgy. Sala, Cabuyao City, Laguna, Philippines. Photograph by Maryan M. Morillo, October 12, 2026.

National Center for Biotechnology Information. (n.d.). PubChem compound summary for sucrose. Retrieved October 12, 2026, from *https://pubchem.ncbi.nlm.nih.gov/compound/Sucrose*

Commission on Isotopic Abundances and Atomic Weights. (2024). Abridged standard atomic weights. *https://www.ciaaw.org/abridged-atomic-weights.htm*

AI use: *AI (Claude) was used to help check the layout of this card and to review my mole-conversion setup. All label values, calculations, references and final answers were independently verified by the student, Maryan M. Morillo.*